

Dominick A. Campbell

PRINCIPAL ENGINEER · AI ARCHITECT · 25+ YEARS · REMOTE

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Summary

Principal Engineer with 25+ years designing systems from Mainframe to modern AI. Shipped \$500M+ in products, led 50+ engineers, saved \$1M annually in cloud costs, and holds 4 patent-pending AI inventions. Currently seeking Principal/Staff roles at companies deploying AI at scale, building data infrastructure, or solving compliance/fintech challenges.

Experience

Vertex Inc

King of Prussia, PA

PRINCIPAL SOFTWARE ENGINEER

May 2022 - May 2026

- Inventor on 4 patent-pending AI systems:

Data Cake: NLP service generating realistic synthetic tax data by learning statistical properties from real customer data using XGBoost, scikit-learn, and deterministic rules engines — enables safe testing and model training without PII exposure

Audit Management Red Black Tree: Self-balancing BST model determining ROI thresholds for audit defense decisions in real-time

Doc Intel: Code/documentation drift confidence scoring

Mort AI Nexus: Economic nexus detection for multi-state sales tax compliance (threshold intelligence: \$100K-\$500K per state)

- Led AI research and architecture for Data & Insights value stream; oversaw launch of Indirect Tax Intelligence and Indirect Tax Close products for 50+ engineers driving Fortune 500 compliance
- Architected data pipelines (WebLogic, Puppet, Linux) processing 50M-100M transactions/day from Target, Starbucks, McDonald's via MS Dynamics, Vertex OSeries, SAP; led metrics overhaul saving ~\$1M/year
- Designed AI Curiosity Workshop ("Raise the Boats") — taught ~50 employees AI fundamentals and agent development
- Creator of Robby — multi-agent SDLC system (70 skills, 15 roles) auto-generating specs, reviews, and sprint plans via MCP integration (ArgoCD, GitHub Actions, Datadog)

JP Morgan Chase

Wilmington, DE

SENIOR SOFTWARE ENGINEER

Apr 2021 - May 2022

- Led Auto initiative data sourcing to single source of truth and AWS architecture migration (Project "Photon")

Connect Your Care

Hunt Valley, MD

SENIOR SOFTWARE ENGINEER

May 2019 - Apr 2021

- Owned HRCommand product — flagship offering that drove United Health/Optum Financial's acquisition of Connect Your Care
- Led systematic replacement of monolithic EJB/J2EE systems with Spring Boot microservices (WebLogic, Puppet) and React frontend
- Managed 5+ member team rewriting legacy systems; designed decomposition strategy, established patterns, mentored team

Donegal Insurance Group

Marietta, PA

SENIOR JAVA/GUIDEWIRE DEVELOPER

Feb 2019 - May 2019

- Converted legacy Insurance Policy system into Guidewire platform as lead developer

TekSystems for JP Morgan Chase

Wilmington, DE

INFRASTRUCTURE DEVELOPER - ADVANCED

Feb 2018 - Feb 2019

- Developed Voice Biometric fingerprinting service (Python, machine learning) for fraud detection resulting in ~\$830B plateau in fraud losses prevented across JPMC
- Pilot program decreased average call throughput by 20 seconds — 25 more customers serviced per rep per day
- Achieved 10 million unique JPMC accounts opted in first 3 months with only 1,300 opting out
- Part of TARA Fraud Busters group developing JPMC CAT tool: aggregated RESTful APIs into unified Angular dashboard with Sapiens rules engine for real-time fraud detection and decision making; built data pipelines processing millions of transaction records daily

Financial Management Solutions

Pasadena, MD

LEAD SOFTWARE DEVELOPER

Apr 2012 - Feb 2018

- Development lead for internal REST Spring Boot/Hibernate service (Java, J2EE, Jenkins, SQLPlus) coordinating IBM AS400 COBOL systems across disparate products
- Led FMS Labor Saver transition from 3rd party vendor to in-house ownership, resulting in \$100,000+/year cost savings
- Oversaw FMS Portal — flagship product delivering sensitive payroll information to enterprise clients handling millions of sensitive records
- AWS Development Lead for FMS Marketing overseeing content promotion and user engagement tracking through Google Analytics

Travelers Insurance

Hunt Valley, MD

LEAD SOFTWARE DEVELOPER

Jun 2010 - Apr 2012

- Led multiple developer teams managing large backlog of issues for Blue Ribbon insurance agent product
- Resolved 200+ issues and completed 1-year estimate in 6 months; continued supporting MVP, LA, and GA development

R2Integrated

Baltimore, MD

SOFTWARE DEVELOPER

Aug 2008 - Jun 2010

- Redesigned Forbes.com Video Network and Embedded Player (Java, J2EE, Jenkins, JDTS, SQLPlus) serving millions of daily users
- Migrated and rewrote code for AAA, Honda, Sirius XM, Pepsi Quaker marketing systems

G1440

Baltimore, MD

SOFTWARE DEVELOPER

Jun 2005 - Jul 2008

- Developer for Sales1440 homebuilding CRM platform (Java, J2EE, Jenkins, JDTS, SQLPlus)

Notable Project & Community

Fantasy Baseball ML Engine — Ensemble ML system with 93+ tests, real-time pipelines, Streamlit UI. github.com/ethos71/fantasy-baseball-ai **Volunteer Coach, Challenger Sports** — Adaptive baseball program for children with special needs, including coaching my 12-year-old son Ethan (autism spectrum).

Patents (Pending)

Data Cake — NLP service generating synthetic tax data using XGBoost and scikit-learn

Audit Management Red Black Tree — Self-balancing BST predicting ROI thresholds for audit defense

Doc Intel — AI system scoring documentation/code drift confidence

Mort AI Nexus — Economic nexus detection for multi-state sales tax compliance

Technical Skills

AI & ML: LLMs, MCP, crewAI, AutoGen, LangChain, Langfuse, Machine Learning, XGBoost, scikit-learn, NLP, PyTorch, SageMaker, Snowflake Cortex, GitHub Copilot, Azure AI, Jupyter, Streamlit, Synthetic Data Generation

Languages: Python, Java (J2EE), JavaScript/TypeScript, C#/.NET, SQL (expert), C++, Go, COBOL, Assembly, PHP

Frameworks: Spring Boot, FastAPI, Django, React, Angular, Hibernate, Microservices, REST APIs, Kafka, Hadoop, Spark

Cloud & Data: AWS (EC2, S3, RDS, Lambda, DynamoDB, Kinesis), Azure, Google Cloud, Linux, Snowflake, PostgreSQL, Oracle (SQLPlus), MongoDB, Redis, Cassandra, DynamoDB

DevOps & Enterprise: Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, CI/CD, WebLogic, Puppet, Configuration Management, Monitoring, Logging, Performance Optimization

Enterprise Systems: JDTS, ITX, Large-scale distributed systems, Secure production deployments, SDLC process management

Education

University of Maryland Baltimore County (UMBC)

BS COMPUTER SCIENCE

2005

AWS CERTIFIED SOLUTIONS ARCHITECT

2018

AWS Re:Invent Bootcamp

GENERATIVE AI FOUNDATIONS

2022